

21-2c Two Extreme Examples of Indifference Curves

The shape of an indifference curve reveals the consumer's willingness to trade one good for the other. When the goods are easy to substitute for each other, the indifference curves are less bowed; when the goods are hard to substitute, the indifference curves are very bowed. To see why this is true, let's consider two extreme cases.

Perfect Substitutes

Suppose that someone offered you bundles of nickels and dimes. How would you rank the different bundles?

Most likely, you would care only about the total monetary value of each bundle. If so, you would always be willing to trade **2** nickels for **1** dime. Your marginal rate of substitution between nickels and dimes would be a fixed number: $MRS = 2$, regardless of the number of nickels and dimes in the bundle.

We can represent your preferences for nickels and dimes with the indifference curves in panel (a) of [Figure 6](#). Because the marginal rate of substitution is constant, the indifference curves are straight lines. In this case of straight indifference curves, we say that the two goods are [perfect substitutes \(two goods with straight-line indifference curves\)](#).

Figure 6

Perfect Substitutes and Perfect Complements

When two goods are perfectly substitutable, such as nickels and dimes, the indifference curves are straight lines, as shown in panel (a). When two goods are perfectly complementary, such as left shoes and right shoes, the indifference curves are right angles, as shown in panel (b).

Two graphs compare (a) perfect substitutes with (b) perfect complements. Graph (a) plotting dimes on the horizontal axis and nickels on the vertical axis, shows three parallel, downward sloping indifference curves, I_1 , I_2 , and I_3 , connecting the points 1, 2, and 3 on the horizontal axis to points 2, 4, and 6 on the vertical axis respectively. Graph (b) plotting right shoes on the horizontal axis and left shoes on the vertical axis, shows two indifference curves, I_1 and I_2 , each shaped like a right angle whose arms run parallel to the axes and whose vertices and the origin are collinear. The vertex of curve I_1 is point (5, 5). Also on curve I_1 , point (7, 5) is marked on the horizontal section and point (5, 7) is marked on the vertical section.

Perfect Complements

Suppose now that someone offered you bundles of shoes. Some of the shoes fit your left foot, others your right foot. How would you rank these different bundles?

In this case, you might care only about the number of pairs of shoes. In other words, you would judge a bundle based on the number of pairs you could assemble from it. A bundle of 5 left shoes and 7 right shoes yields only 5 pairs. Getting 1 more right shoe has no value if there is no left shoe to go with it.

We can represent your preferences for right and left shoes with the indifference curves in panel (b) of [Figure 6](#). In this case, a bundle with 5 left shoes and 5 right shoes is just as good as a bundle with 5 left shoes and 7 right shoes. It is also just as good as a bundle with 7 left shoes and 5 right shoes. The indifference curves, therefore, are right angles. In this case of right-angle indifference curves, we say that the two goods are [perfect complements \(two goods with right-angle indifference curves\)](#).

In the real world, most goods are neither perfect substitutes (like nickels and dimes) nor perfect complements (like right shoes and left shoes). Perfect substitutes and perfect complements are extreme cases. They are introduced here not because they are common but because they illustrate how indifference curves reflect a consumer's preferences. For most goods, the indifference curves are bowed inward, but not so bowed that they become right angles.

QuickQuiz

3. At two points on an indifference curve,
 - a. the consumer has the same income.
 - b. the consumer has the same marginal rate of substitution.
 - c. the bundles of goods cost the consumer the same amount.
 - d. the bundles of goods yield the consumer the same satisfaction.
4. At any point on an indifference curve, the slope of the curve measures the consumer's
 - a. income.
 - b. willingness to trade one good for the other.
 - c. perception of the two goods as substitutes or complements.
 - d. elasticity of demand.

Chapter 21: The Theory of Consumer Choice: 21-2c Two Extreme Examples of Indifference Curves

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